SQL ANALYSIS

1: Total Customers

Objective: Determine the total number of customers in the dataset.

```
SELECT COUNT(*) AS total_customers

FROM churn_data;
```

Result

Result Grid | Filter Ro

	total_customers
▶	7043

Insight: This represents the total active and churned customer base of the telecom company.
All further churn analysis is based on this population.

Business Value: Understanding the total customer base helps measure customer retention and churn rate accurately.

2: Churn Distribution

Objective: Identify how many customers have churned vs retained.

```
SELECT

Churn,

COUNT(*) AS customer_count

FROM churn_data

GROUP BY Churn;
```

	Churn	customer_count
▶	No	5174
	Yes	1869

Insight: A significant portion of customers have churned, which indicates **customer retention challenges**.

Business Value: This helps companies evaluate **customer retention performance**.

3: Churn Rate

Objective: Calculate the percentage of customers who churned.

```
SELECT
ROUND(
SUM(CASE WHEN TRIM(Churn) = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*),
2
) AS churn_rate_percentage
FROM churn_data;
```

	churn_rate_percentage
▶	26.54

Insight: About one quarter of customers leave the service, which is a high churn rate for telecom companies.

Business Value: Reducing churn even by 5% can significantly increase revenue.

4: Churn by Contract Type

Objective: Analyze how contract type influences churn.

```
SELECT
Contract,
Churn,
COUNT(*) AS customers
FROM churn_data
GROUP BY Contract, Churn;
```

	Contract	Churn	customers
▶	Month-to-month	No	2220
	One year	No	1307
	Month-to-month	Yes	1655
	Two year	No	1647
	Two year	Yes	48
	One year	Yes	166

Insight: Customers with month-to-month contracts show the highest churn rate, while two-year contracts show the lowest churn.

Business Value: Encouraging customers to choose longer-term contracts can improve retention.

5: Churn by Internet Service

Objective: Understand whether internet service type affects churn.

```
SELECT

InternetService,

Churn,

COUNT(*) AS customers

FROM churn_data

GROUP BY InternetService, Churn;
```

	InternetService	Churn	customers
▶	DSL	No	1962
	DSL	Yes	459
	Fiber optic	Yes	1297
	Fiber optic	No	1799
	No	No	1413
	No	Yes	113

Insight: Customers using fiber optic services tend to churn more frequently.

Business Value: This may indicate service quality or pricing issues with fiber plans.

6: Churn by Payment Method

Objective: Analyze how payment methods influence churn.

```
SELECT

PaymentMethod,

Churn,

COUNT(*) AS customers

FROM churn_data

GROUP BY PaymentMethod, Churn;
```

	PaymentMethod	Churn	customers
▶	Electronic check	No	1294
	Mailed check	No	1304
	Mailed check	Yes	308
	Bank transfer (automatic)	No	1286
	Electronic check	Yes	1071
	Credit card (automatic)	No	1290
	Bank transfer (automatic)	Yes	258
	Credit card (automatic)	Yes	232

1290

Insight: Customers paying via electronic check show the highest churn rate.

Business Value: This could indicate less commitment compared to automated payment methods.

Encouraging automatic payments could reduce churn.

7: Average Monthly Charges by Churn

Objective: Analyze whether higher charges influence churn.

```
SELECT

Churn,

AVG(MonthlyCharges) AS avg_monthly_charges

FROM churn_data

GROUP BY Churn;
```

Churn	avg_monthly_charges
No	61.265124
Yes	74.441332

Insight: Customers who churn tend to have higher monthly charges on average.

Business Value: Price-sensitive customers may leave due to high subscription costs.

8: Revenue Loss Due to Churn

Objective: Estimate monthly revenue lost from churned customers.

```
SELECT

SUM(MonthlyCharges) AS monthly_revenue_lost

FROM churn_data

WHERE Churn='Yes';
```

monthly_revenue_lost
139130.85

Insight: Churned customers represent significant lost monthly revenue.

Business Value: Reducing churn can directly increase company revenue.

9: High Value Customers

Objective: Identify the highest revenue generating customers.

```
SELECT

customerID,
```

```
MonthlyCharges,
TotalCharges
FROM churn_data
ORDER BY TotalCharges DESC
LIMIT 10;
```

customerID	MonthlyCharges	TotalCharges
2889-FPWRM	117.80	8684.80
7569-NMZYQ	118.75	8672.45
9739-JLPQJ	117.50	8670.10
9788-HNGUT	116.95	8594.40
8879-XUAHX	116.25	8564.75
9924-JPRMC	118.20	8547.15
0675-NCDYU	116.40	8543.25
6650-BWFRT	117.15	8529.50
0164-APGRB	114.90	8496.70
1488-PBLJN	116.85	8477.70

Insight: These customers represent the **most valuable customer segment**.

Business Value: Companies should prioritize **retention strategies for high-value customers**.

10: Churn by Tenure

Objective: Understand churn behavior based on customer tenure.

```
SELECT
tenure,
Churn,
COUNT(*) AS customers
FROM churn_data
GROUP BY tenure, Churn;
```

tenure	Churn	customers
1	No	233
34	No	53
2	Yes	123
45	No	55
8	Yes	42
22	No	63
10	No	71
28	Yes	12
62	No	65
13	No	71
16	No	52
58	No	56
49	Yes	15

Result 32

Insight: Customers with shorter tenure tend to churn more frequently.

Business Value: Improving customer onboarding and early engagement can reduce churn.

Key Findings:

High churn rate among month-to-month contract users

Customers with higher monthly charges churn more

Electronic check users show higher churn

New customers are more likely to churn

Recommendations:

Encourage long-term contracts

Introduce loyalty programs

Offer discounts for automatic payments

Improve early customer engagement